

CAPE ROMANZOF, AK, USA

WMO: 702120

Lat: 61.783N

Lon: 166.033W

Elev: 146

StdP: 99.58

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26646

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-23.3	-28.7	0.3	-24.3	-27.1	0.3	-22.2	25.3	-14.2	23.3	-16.6	9.5	10	0.999

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.1	18.5	13.5	16.7	12.7	14.9	11.9	14.4	17.4	13.2	15.8	12.3	14.4	4.9	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.5	15.0	12.0	8.9	14.1	11.2	8.4	13.3	40.6	17.4	37.6	15.9	35.3	14.4	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.8	18.2	16.1	DB	-24.8	21.4	3.0	2.5	-27.0	23.2	-28.7	24.6	-30.5	26.0	-32.7	27.8
(5)				WB	-25.0	15.5	3.0	1.7	-27.1	16.7	-28.9	17.7	-30.5	18.7	-32.7	19.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-10.5	-8.9	-9.2	-3.9	2.7	7.9	10.6	10.4	7.4	1.5	-4.5	-7.7	(6)
(7)		DBStd	9.33	8.09	7.43	6.64	5.64	4.34	3.61	2.84	2.74	2.88	3.57	4.92	6.23	(7)
(8)		HDD10.0	3874	637	529	596	417	229	82	25	26	85	264	434	550	(8)
(9)		HDD18.3	6806	896	763	854	667	484	312	242	247	327	523	684	808	(9)
(10)		CDD10.0	112	0	0	0	0	3	20	42	38	8	0	0	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)
(12)		CDH23.3	3	0	0	0	0	0	0	3	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.3	9.3	9.1	8.5	7.7	6.3	5.2	5.1	5.5	6.7	7.4	8.4	8.9	(14)
(15)	Precipitation	PrecAvg	714	49	48	34	30	38	54	77	98	96	71	66	53	(15)
(16)		PrecMax	974	110	117	107	60	82	82	117	147	170	142	130	111	(16)
(17)		PrecMin	573	10	14	4	7	4	24	46	34	14	20	13	12	(17)
(18)		PrecStd	94	24	26	22	13	18	15	21	26	29	25	30	27	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	3.9	3.5	8.5	17.1	20.3	22.9	20.8	16.3	10.5	6.9	4.9	(19)
(20)		MCWB	1.9	1.3	0.9	4.3	10.3	13.2	16.3	15.0	11.4	7.5	5.5	2.2	2.2	(20)
(21)		2%	DB	3.4	2.0	2.0	6.3	13.2	17.1	18.7	17.9	13.5	8.4	4.6	2.8	(21)
(22)		MCWB	1.0	0.3	0.1	3.3	8.2	11.9	13.5	14.0	10.9	7.1	3.4	1.4	1.4	(22)
(23)		5%	DB	1.4	0.9	0.8	4.8	10.7	15.0	16.6	16.0	12.1	7.4	3.1	1.3	(23)
(24)		MCWB	0.2	-0.2	-0.3	2.4	6.9	11.0	13.0	13.3	10.3	6.4	2.4	0.3	0.3	(24)
(25)		10%	DB	0.1	0.1	-0.4	3.3	8.8	13.2	14.7	14.1	10.9	6.2	2.1	0.2	(25)
(26)		MCWB	-0.7	-0.7	-1.3	1.4	5.9	10.2	12.0	12.4	9.5	5.1	1.5	-0.5	-0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.0	1.8	4.6	11.0	14.0	16.2	16.3	12.5	8.5	5.7	3.4	(27)
(28)		MADB	5.1	3.0	2.9	7.7	15.4	18.6	22.5	18.9	14.6	9.4	6.7	3.8	3.8	(28)
(29)		2%	WB	1.6	0.8	0.6	3.6	8.9	12.6	14.4	14.6	11.5	7.4	3.7	1.6	(29)
(30)		MADB	2.8	1.4	1.4	6.0	12.3	16.0	17.8	17.1	12.9	8.1	4.3	2.5	2.5	(30)
(31)		5%	WB	0.4	0.1	-0.2	2.5	7.4	11.5	13.2	13.5	10.5	6.4	2.5	0.6	(31)
(32)		MADB	1.2	0.7	0.6	4.5	10.3	14.4	15.9	15.6	11.9	7.4	3.0	1.2	1.2	(32)
(33)		10%	WB	-0.8	-0.6	-1.2	1.6	6.1	10.4	12.3	12.5	9.8	5.4	1.4	-0.3	(33)
(34)		MADB	0.1	0.0	-0.5	3.2	8.5	13.0	14.5	14.0	10.9	6.1	1.9	0.3	0.3	(34)
(35)	Mean Daily Temperature Range	MDBR	5.4	5.5	5.4	5.0	4.7	5.0	4.1	3.3	3.1	3.1	4.1	4.9	(35)	
(36)		5% DB	MCDBR	6.2	5.1	5.4	5.4	7.5	7.7	7.2	5.8	4.4	3.2	3.7	4.9	(36)
(37)		MCWBR	5.1	4.4	4.5	3.6	4.8	4.9	4.5	3.9	3.2	2.9	3.3	4.1	(37)	
(38)		5% WB	MCDBR	5.9	4.9	4.9	5.2	6.9	7.2	6.2	5.2	3.9	3.1	3.7	4.1	(38)
(39)		MCWBR	5.2	4.5	4.3	3.7	4.7	4.8	4.2	3.7	3.1	3.1	3.5	3.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.212	0.277	0.282	0.345	0.364	0.364	0.370	0.383	0.335	0.300	0.250	0.203	(40)	
(41)		taud	2.035	2.122	2.200	2.061	2.215	2.414	2.441	2.370	2.476	2.358	2.032	1.879	(41)	
(42)		Ebn at Noon	496	679	833	832	851	862	846	797	778	662	408	279	(42)	
(43)		Edh at Noon	36	70	92	132	123	104	99	98	73	56	37	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.30	1.04	2.77	4.47	5.04	5.16	4.19	3.30	2.14	1.04	0.32	0.13	(44)	
(45)		RadStd	0.04	0.13	0.33	0.40	0.60	0.47	0.30	0.42	0.19	0.08	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page